

HARSH SHROFF

Open to Relocation

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SUMMARY

AWS Machine Learning Associate Certified AI/ML Engineer and Data Scientist with hands-on experience delivering applied AI systems across sectors including geospatial intelligence, robotics, and precision aquaculture. Specializing in generative AI, LLMops, and agentic AI architectures, with a portfolio of production-ready tools built using LangChain, Claude, YOLOv8, Vertex AI, and Streamlit. Proven success in automating unstructured data pipelines, deploying real-time CV/NLP solutions, and co-authoring peer-reviewed research. Known for bridging the gap between experimentation and execution—translating complex models into scalable, user-facing platforms that drive real-world impact. Ready to lead AI transformation efforts with a rare blend of deep technical fluency, strategic thinking, and an outcome-first mindset.

SKILLS

Machine Learning & AI: Generative AI (*LLMs, RAG (Retrieval-Augmented Generation), LangChain, LlamaIndex*), Deep Learning (*TensorFlow, PyTorch, YOLOv8, OpenCV*), Natural Language Processing (*NLP*), Computer Vision (*CV*), Time Series Forecasting, Predictive Modeling, MLOps (*MLflow, Kubeflow*), Prompt Engineering

Data Engineering & Cloud Computing: AWS (*Lambda, EC2, S3, Textract, Comprehend, Glue, Redshift, Kinesis, Bedrock*), SQL & NoSQL Databases (*PostgreSQL, MySQL, MongoDB, Elasticsearch*), Data Pipelines (*ETL, Airflow, Terraform, Spark, Hadoop*)

Data Analysis & Visualization: Python (*Pandas, NumPy, SciPy, Seaborn, Matplotlib, plotly*), BI Tools (*Tableau, Power BI, Streamlit, Dash*), Statistical Analysis (*A/B Testing, Regression, Hypothesis Testing*), CI/CD, Git

Software Development & Automation: Python, R, SQL, C++, Embedded C, API Development (*FastAPI, Flask*), DevOps (*CI/CD, Docker, Kubernetes, Terraform*), Agile Methodologies, Version Control

EXPERIENCE

ML Engineer

Jun 2024 – May 2025

VITG Corp., Halethorpe, MD

- Developed a resume scoring Slack bot powered by Llama LLM and AWS services (Lambda, EC2, S3) to automate candidate screening, reducing manual processing by 60%.
- Built a chatbot-driven geospatial tool to match addresses with government conservation zones, aiding environmental planning and property regulations in Hyderabad.
- Automated geospatial data extraction from 2500+ scanned PDFs/images using OCR-based lat-long extraction and Claude 3.5 Haiku LLM, visualizing results on interactive maps.
- Designed and deployed an AI-powered web-app using Streamlit and AWS for real-time geospatial analysis, improving data accessibility by 30%.

AI Researcher

Mar 2023 – Dec 2024

Center for Real-time Distributed Sensing and Autonomy, UMBC

- Led a team of 5 to develop AI-powered multi-modal robotic navigation for Boston Dynamics Spot, Clearpath Husky & Jackal robots under the ArtIAMAS project (funded by the US Army Research Lab & DEVCOM).
- Implemented gesture-based robotic controls using Microsoft HoloLens 2 and Python, enabling intuitive human-robot interaction for the "Human Machine Teaming" project.
- Optimized LIDAR-based pathfinding algorithms using reinforcement learning, reducing navigation errors by 20% in dynamic environments.
- Designed a GPS-based navigation system for Clearpath Husky robots, achieving 95% path accuracy.

Data Science Intern

Aug 2023 – Dec 2023

Freshwater Institute – The Conservation Fund, Shepherdstown, WV

- Achieved a 92% accuracy rate in fish fillet color grading using Python, OpenCV & YOLOv8 Classifier model deployed on Raspberry Pi 4.
- Designed and developed an in-house 3D printed portable case for a real-time assessment on fillet grading in aquaculture.
- Co-authored a research paper titled "Fish Fillet Color Profiling Using Deep Learning", detailing the application of AI in precision aquaculture and food safety.

EDUCATION

Master's in Data Science

May 2024

University of Maryland Baltimore County (UMBC)

- GPA: 3.8/4.0
- Relevant Coursework: Natural Language Processing, Data Analysis & Visualization, Data Management, Big Data Processing, Financial Data Science, Machine Learning, Artificial Intelligence

Bachelor's in ELectronics & Communication Engineering

May 2022

Gujarat Technological University (GTU)

- GPA: 3.8/4.0 – *converted*
- Relevant Coursework: Introduction to Python, Introduction Machine Learning & Artificial Intelligence

PROJECTS

Document Parsing for Environmental Planning

AWS (Textract, S3, Redshift), Python, Tableau

- Automated geospatial data extraction from 2500+ scanned PDFs using OCR and ML models, increasing data processing efficiency by 40%.
- Developed prompt engineering strategies to refine AI outputs for conservation project prioritization.
- Designed interactive Streamlit web-app, providing stakeholders with real-time insights, and enabling faster decision-making on conservation projects.

Dashboard for Autonomous Robotic Patrols

Python, ROS, LiDAR, Deep Learning, Computer Vision

- Developed a real-time asset monitoring dashboard using Python, Dash and ROS, tracking battery health, network connectivity, and asset movement via GPS in military simulations.
- Collaborated with cross-functional teams to define use cases and blended outputs from various systems into one.
- Integrated real-time object detection using OpenCV and Python and obstacle detection via LiDAR, enhancing robotic autonomous movement and safety.

AirbnbLense – Personalized Recommendation System ([GitHub](#))

Python, OpenAI API, Computer Vision, Tableau, Big Data

- Built a recommendation engine combining structured and unstructured data (160MBs of CSV, 130GBs of images).
- Implemented computer vision models to analyze listing images for style, color, and object recognition.
- Reduced Airbnb bounce rates by 20% and improved engagement by 40% through tailored recommendations.

Cyclists' Performance Dashboard ([GitHub](#))

Python (Dash, Seaborn), Machine Learning

- Developed an interactive dashboard to track real-time cyclist performance metrics using Python and Dash, improving training efficiency by 20%.
- Implemented predictive analytics for personalized training insights, enhancing athlete performance.

ACHIEVEMENTS

Certifications

- **AWS Machine Learning Associate Certification** – [View Badge](#)
- **AWS Machine Learning Foundations Certification** – Proficiency in core ML concepts & AWS tools.

Publications

- **Ranjan, R., Shroff, H., Sharrer, K., Tsukuda, S., & Good, C.** (2024). *FilletCam AI: A handheld tool for precise fillet color profiling of Atlantic salmon and rainbow trout*. *Journal of Agriculture and Food Research*, 18, 101461. <https://doi.org/10.1016/j.jafr.2024.101461>
- **Trivedi, K., & Shroff, H.** (2021). *Identification of deadliest mosquitoes using wing beats sound classification on tiny embedded system using machine learning and Edge Impulse platform*. In *2021 ITU Kaleidoscope: Connecting Physical and Virtual Worlds (ITU K)* (pp. 1–6). IEEE. <https://doi.org/10.23919/ITUK53220.2021.9662116>